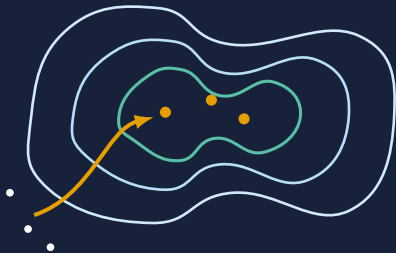
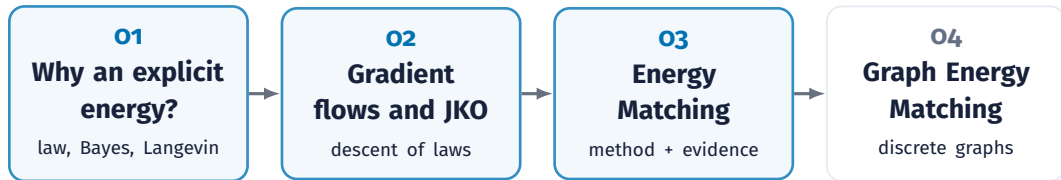


Energy Matching

One potential for transport
and equilibrium

Michał Balcerak • University of Zurich / ETH AI Center
Starkly Speaking Reading Group • 13 July 2026





"If you want a new idea, read an old book."

— anonymous maxim, often attributed to Ivan Pavlov

THE TWO PAPERS BEHIND THIS TALK

Energy Matching: Unifying Flow Matching and Energy-Based Models for Generative Modeling

continuous spaces
NeurIPS 2025

Michal Balcerak¹, Tamaz Amiranashvili^{1,2}, Antonio Terpin³, Suprosanna Shit¹
Lea Bogensperger¹, Sebastian Kaltenbach⁴, Petros Koumoutsakos⁴, Bjoern Menze¹

arXiv:2504.10612

Graph Energy Matching: Transport-Aligned Energy-Based Modeling for Graph Generation

discrete graphs
2026 preprint

Michal Balcerak¹, Suprosanna Shit¹, Chinmay Prabhakar¹, Sebastian Kaltenbach⁴
Michael S. Albergo^{4,5}, Yilun Du^{4,5,*}, Bjoern Menze^{1,*}

arXiv:2603.23398

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⁴ Harvard University ⁵ Kempner Institute * equal senior contribution

The unknown data law is the learnable object

PROBABILITY AS THE OBJECT

A dataset gives finite observations of an unknown continuous law:

$$x_1, \dots, x_n \stackrel{\text{iid}}{\sim} p_{\text{data}}(x), \quad x \in \mathbb{R}^d.$$

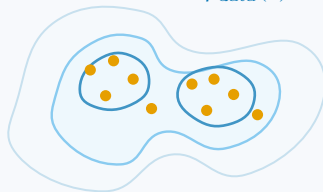
The modelling target is the law itself: $p_\theta \approx p_{\text{data}}$.

This remains central after conditioning. Bayes' rule gives

$$\underbrace{p(x | y)}_{\text{inverse problem}} \propto \underbrace{p(y | x)}_{\text{forward model}} \underbrace{p_{\text{data}}(x)}_{\text{learned data prior}}.$$

The forward model is often known or easier. The learned prior restricts solutions to plausible data, so the difficult reusable object is still $p_{\text{data}}(x)$.

the unknown law $p_{\text{data}}(x)$



the dataset gives finitely many probes of this field

Sampling models hide likelihoods. An EBM exposes energy

ITERATIVE ACCESS VS. ONE SCALAR

A sampling route

Many modern generative models are accessed by repeatedly applying a learned process:

noise → network → network → sample

This produces a sample, but relative likelihoods are not a direct network output.

An exposed scalar field

An energy-based model returns

$$x \xrightarrow{E_\theta} E_\theta(x), \quad p_\theta(x) = \frac{e^{-E_\theta(x)}}{Z_\theta}.$$

Evaluating the scalar at x and x' exposes their relative likelihood directly:

$$\log \frac{p_\theta(x)}{p_\theta(x')} = -(E_\theta(x) - E_\theta(x')).$$

Z_θ blocks normalised likelihood evaluation, but cancels exactly in likelihood ratios.

An energy also defines a Langevin sampler

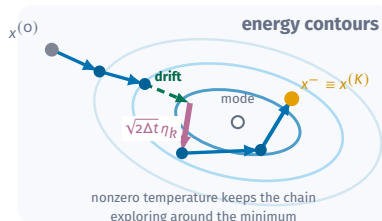
ENERGY $\Rightarrow$ SAMPLING $\Rightarrow$ CONTRASTIVE LEARNING

For a smooth confining energy,

$$p_{\theta}(x) = \frac{e^{-E_{\theta}(x)}}{Z_{\theta}}, \quad dX_t = -\nabla E_{\theta}(X_t) dt + \sqrt{2} dW_t.$$

Euler-Maruyama with step Δt gives

$$x^{(k+1)} = x^{(k)} - \Delta t \nabla E_{\theta}(x^{(k)}) + \sqrt{2\Delta t} \eta_k, \quad \eta_k \sim \mathcal{N}(\mathbf{o}, I_d).$$



Maximum likelihood has a contrastive gradient

$$\begin{aligned} \mathcal{L}_{\text{ML}} &:= -\mathbb{E}_{p_{\text{data}}} \log p_{\theta} = \mathbb{E}_{p_{\text{data}}} [E_{\theta}] + \log Z_{\theta}, \\ \nabla_{\theta} \mathcal{L}_{\text{ML}} &= \mathbb{E}_{p_{\text{data}}} [\nabla_{\theta} E_{\theta}] - \mathbb{E}_{p_{\theta}} [\nabla_{\theta} E_{\theta}] \\ &\approx \mathbb{E}_{x^{+}} [\nabla_{\theta} E_{\theta}(x^{+})] - \mathbb{E}_{x^{-}} [\nabla_{\theta} E_{\theta}(x^{-})]. \end{aligned}$$

Endpoints approximate the model expectation.

Poorly mixed finite chains constrain only the regions they reach.

The density target gives no finite off-data landscape

WHAT DOES $-\log p$ SAY AWAY FROM DATA?

Where the continuous density is positive, its ideal energy is

$$E^*(x) = -\log p_{\text{data}}(x) + C, \quad p_{\text{data}}(x) > 0.$$

At a structural zero,

$$p_{\text{data}}(x) = 0 \implies E^*(x) = +\infty.$$

Thus the exact density target does not provide a finite, traversable continuation away from data.

Yet generation commonly starts from noise, precisely where this ideal density energy provides no usable gradient.



Noise-initialised samples may start off data, where density does not specify a finite continuation.

Density fitting constrains the field **on or near data**. Sampling needs a useful field globally.

Density energy is not automatically a transport landscape

THE DESIGN BRIEF

Local density, global traps



Finite training can leave off-data wells and barriers. Langevin chains can become trapped.

On / near data

$$V_{\theta}(x) \approx -\log p_{\text{data}}(x) + C$$

Transport-aware continuation



Preserve the local density wells while making the global landscape easier to traverse.

Away from data

$-\nabla V_{\theta}(x)$: a coherent route from noise

Can **one scalar** encode both equilibrium and a route? $\longrightarrow$ Gradient flows and JKO.

Start with ordinary gradient flow

DESCENT OF A POINT

For $F : \mathbb{R}^d \rightarrow \mathbb{R}$, ordinary gradient flow is

$$\dot{x}_t = -\nabla F(x_t), \quad \frac{d}{dt} F(x_t) = -\|\nabla F(x_t)\|^2 \leq 0.$$

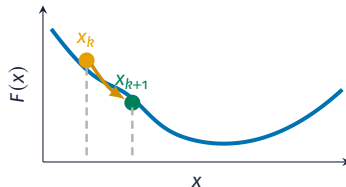
One variational time step is

$$x_{k+1} \in \arg \min_x \left\{ F(x) + \frac{1}{2\Delta t} \|x - x_k\|^2 \right\}.$$

First-order optimality at a differentiable minimiser gives

$$\begin{aligned} 0 &= \nabla_x \left[F(x) + \frac{1}{2\Delta t} \|x - x_k\|^2 \right] \Big|_{x=x_{k+1}} \\ &= \nabla F(x_{k+1}) + \frac{x_{k+1} - x_k}{\Delta t}, \\ \iff \quad &\boxed{\frac{x_{k+1} - x_k}{\Delta t} = -\nabla F(x_{k+1})} \quad \text{implicit Euler.} \end{aligned}$$

JKO keeps this variational template and changes only the **state** and the **metric**.



Lower energy, but movement is penalised.

Optimal transport supplies a metric between laws

DESCENT OF A DISTRIBUTION

Now the state is a probability law $\rho \in \mathcal{P}_2(\mathbb{R}^d)$. Write $x \sim \rho_0$ for the source and $y \sim \rho_1$ for the target. For the squared Euclidean ground cost,

$$W_2^2(\rho_0, \rho_1) = \inf_{\gamma \in \Gamma(\rho_0, \rho_1)} \int \underbrace{\|x - y\|_2^2}_{c(x,y)} \gamma(dx, dy).$$

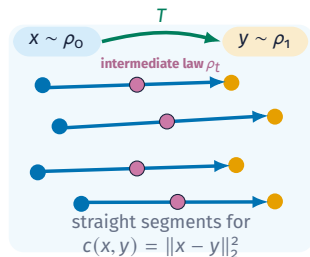
A coupling selects the pairing. In practice, we approximate it on a minibatch and use the resulting transport map T .

The solver gives a map

$$C_{ij} = c(x_i, y_j), \quad \pi^* \in \arg \min_{\pi \in S_B} \sum_{i=1}^B C_{i, \pi(i)},$$

$$T(x_i) = y_{\pi^*(i)}, \quad T_{\#}\rho_0 = \rho_1.$$

For the rest of the talk, treat T as given.



The map gives a path of laws

$$T_t = (1 - t) \text{Id} + tT, \quad \rho_t = (T_t)_{\#}\rho_0, \quad t \in [0, 1],$$
$$x_t = T_t(x) = (1 - t)x + tT(x), \quad y = T(x).$$

For the squared Euclidean cost, x_t is a straight line. Thus ρ_t is the displacement interpolation between the laws.

Why was the JKO scheme introduced?

THE 1998 FOKKER–PLANCK QUESTION

The **Jordan–Kinderlehrer–Otto (JKO) scheme** began with particles driven by a potential force and Brownian motion. Their law follows the Fokker–Planck equation:

$$dX_t = -\nabla\Psi(X_t) dt + \sqrt{2\beta^{-1}} dW_t \implies \partial_t \rho_t = \nabla \cdot (\rho_t \nabla \Psi) + \beta^{-1} \Delta \rho_t.$$

The question

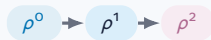
Can this dissipative PDE be understood as **steepest descent**, just like ordinary gradient flow?

The free energy

$$\mathcal{F}(\rho) = \int \Psi d\rho + \beta^{-1} \int \rho \log \rho dx.$$

The potential force concentrates particles while Brownian motion spreads them.

Their construction



Each implicit step balances lower free energy against Wasserstein movement.

Their result: Fokker–Planck dynamics are steepest descent of free energy in the Wasserstein metric. The discrete variational iterates converge to the PDE as $\Delta t \rightarrow 0^+$.

R. Jordan, D. Kinderlehrer and F. Otto, *The Variational Formulation of the Fokker–Planck Equation*, 1998.

JKO is implicit Euler in probability space

THE VARIATIONAL STEP

$$\rho_{k+1} \in \arg \min_{\rho \in \mathcal{P}_2(\mathbb{R}^d)} \left\{ \mathcal{F}(\rho) + \frac{1}{2\Delta t} W_2^2(\rho_k, \rho) \right\}.$$

Euclidean proximal descent

state x_k
energy $F(x)$
distance $\|x - x_k\|^2$

Wasserstein proximal descent

state ρ_k
free energy $\mathcal{F}(\rho)$
distance $W_2^2(\rho_k, \rho)$

Optimality of the JKO update yields the discrete energy-dissipation inequality:

$$\mathcal{F}(\rho_{k+1}) + \frac{1}{2\Delta t} W_2^2(\rho_k, \rho_{k+1}) \leq \mathcal{F}(\rho_k).$$

Potential compresses and entropy spreads

TWO COMPETING EFFECTS

Use the free energy

$$\mathcal{F}_\varepsilon(\rho) = \underbrace{\mathbb{E}_{X \sim \rho}[V(X)]}_{\text{prefer low potential}} - \underbrace{\varepsilon H(\rho)}_{\text{reward spread}}, \quad H(\rho) = - \int \rho(x) \log \rho(x) \, dx.$$

Potential term: concentrate



Lowering $\mathbb{E}_\rho[V]$ moves mass toward minima of V .

Entropy term: resist collapse



Lowering $-\varepsilon H(\rho)$ rewards spread.

$$X_{k+1} = X_k - \Delta t \nabla V(X_k) + \sqrt{2\varepsilon \Delta t} \eta_k, \quad \eta_k \sim \mathcal{N}(\mathbf{0}, I_d).$$

The drift compresses toward low energy while Gaussian noise spreads. For a confining V , they can balance at equilibrium.

Perturbations reveal the JKO first-order condition

A CHARACTERISTIC OF THE SOLUTION

Let $\rho^+ = \rho_{k+1}$ be a smooth JKO minimiser and let $S_{\#}\rho^+ = \rho_k$, so the new point y is paired with $x = S(y)$. Perturb the **new law** along any smooth vector field ξ :

$$\rho_s = (\text{id} + s\xi)_{\#}\rho^+.$$

Differentiate the JKO objective at the minimiser

$$\begin{aligned} 0 &= \left. \frac{d}{ds} \right|_{s=0} \left[\mathcal{F}_{\varepsilon}(\rho_s) + \frac{1}{2\Delta t} W_2^2(\rho_k, \rho_s) \right] \\ &= \int \left\langle \underbrace{\frac{y - S(y)}{\Delta t}}_{\text{movement variation}} + \underbrace{\nabla V(y) + \varepsilon \nabla \log \rho^+(y)}_{\nabla \delta \mathcal{F}_{\varepsilon} / \delta \rho}, \xi(y) \right\rangle \rho^+(dy). \end{aligned}$$

This holds for every ξ , so the bracket vanishes ρ^+ -almost everywhere. With $x = S(y)$, equivalently $y = T(x)$:

$$\frac{y - x}{\Delta t} = -\nabla V(y) - \varepsilon \nabla \log \rho^+(y).$$

This is implicit Euler: the force is evaluated at the new point y and the new law ρ^+ .

One JKO step connects transport to an EBM

TRANSPORT AND EQUILIBRIUM

Let $x \sim \rho_t$ be a source point under the old law and let $y = T(x) \sim \rho_{t+\Delta t}$ be its paired target under the new law. The condition from the previous slide reads

$$\underbrace{\frac{y-x}{\Delta t}}_{\text{transport velocity}} + \underbrace{\nabla V(y)}_{\text{potential gradient}} + \underbrace{\varepsilon \nabla \log \rho_{t+\Delta t}(y)}_{\text{entropy term}} = 0.$$

Zero temperature: movement

If $\varepsilon = 0$,

$$\frac{y-x}{\Delta t} = -\nabla V(y).$$

Deterministic transport.

Equilibrium: an EBM

At equilibrium the optimal coupling is the identity, so $x = y$:

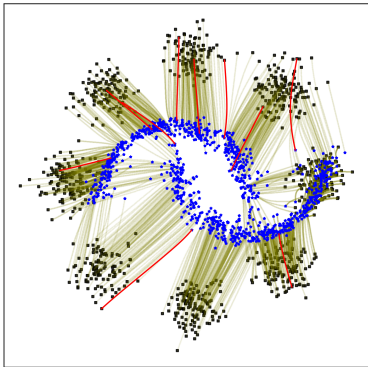
$$\nabla V + \varepsilon \nabla \log \rho_{\text{eq}} = 0 \implies \rho_{\text{eq}} \propto e^{-V/\varepsilon}.$$

Can we **learn one time-independent** V_θ that is transport-like first and density-correct at equilibrium?

Eight Gaussians → two moons

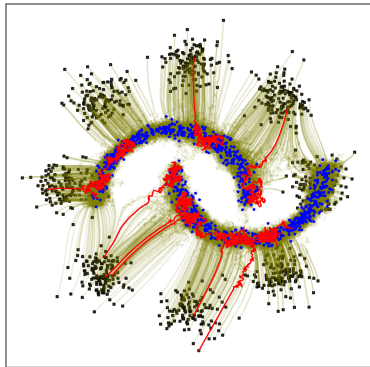
THE WHOLE IDEA IN ONE PICTURE

OT-Flow Matching



$v_\theta(x, t) \in \mathbb{R}^d$: a time-indexed route to the target.

Energy Matching



$V_\theta(x) \in \mathbb{R}$: directed arrival, then continued exploration.

black: eight-Gaussian source • blue: two-moons target • olive: trajectories • red: selected paths

Energy Matching: one potential, two regimes

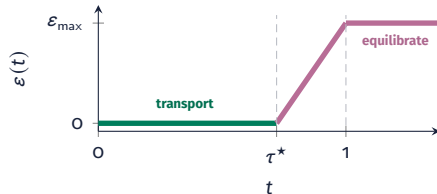
A TIME-INDEPENDENT SCALAR, A TIME-DEPENDENT TEMPERATURE

Learn one scalar $V_\theta : \mathbb{R}^d \rightarrow \mathbb{R}$, and vary only the entropy coefficient:

$$\varepsilon(t) = \begin{cases} 0, & 0 \leq t < \tau^\star, \\ \varepsilon_{\max} \frac{t - \tau^\star}{1 - \tau^\star}, & \tau^\star \leq t < 1, \\ \varepsilon_{\max}, & t \geq 1. \end{cases}$$

The resulting dynamics are always

$$dX_t = -\nabla V_\theta(X_t) dt + \sqrt{2\varepsilon(t)} dW_t.$$



At late time

$$\rho_{\text{eq}}(x) \propto \exp\left[-\frac{V_\theta(x)}{\varepsilon_{\max}}\right], \quad E_\theta^{\text{EBM}} = \frac{V_\theta}{\varepsilon_{\max}}.$$

Far from data, V_θ supplies a route. Near equilibrium, the same V_θ supplies the density.

Regime I: match the global transport field

THE $\mathcal{L}_{\text{OT}}$ OBJECTIVE

Match a Gaussian minibatch to a data minibatch by quadratic OT.
The solver returns

$$z \sim \rho_0, \quad T(z) \sim p_{\text{data}}.$$

Interpolate along the paired segment:

$$x_t = (1-t)z + tT(z), \quad \dot{x}_t = T(z) - z.$$

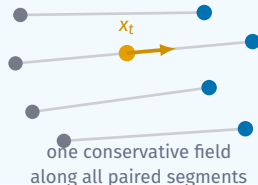
Since the model velocity is $-\nabla V_\theta$, train

$$\mathcal{L}_{\text{OT}} = \mathbb{E}_{\substack{z \sim \rho_0 \\ t \sim U(0, \tau^*)}} \|\nabla V_\theta(x_t) + T(z) - z\|^2.$$

This is **OT-like** velocity regression under the restrictive ansatz $v(x) = -\nabla V_\theta(x)$: time-independent and irrotational.

Gaussian noise

data



Regime II: match the equilibrium density

THE $\mathcal{L}_{\text{CL}}$ OBJECTIVE

Near equilibrium, the target is the Boltzmann law

$$p_{\text{eq}}(x) \propto \exp\left[-\frac{V_{\theta}(x)}{\varepsilon_{\text{max}}}\right].$$

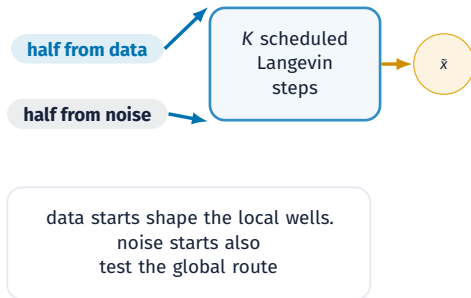
Use the contrastive loss directly:

$$\mathcal{L}_{\text{CL}} = \mathbb{E}_{x \sim p_{\text{data}}}\left[\frac{V_{\theta}(x)}{\varepsilon_{\text{max}}}\right] - \mathbb{E}_{\tilde{x} \sim \text{sg}(p_{\text{eq}})}\left[\frac{V_{\theta}(\tilde{x})}{\varepsilon_{\text{max}}}\right].$$

Negative samples are approximated by the same scheduled Langevin chain:

$$x^{m+1} = x^m - \Delta t \nabla V_{\theta}(x^m) + \sqrt{2\Delta t \varepsilon(m\Delta t)} \eta^m.$$

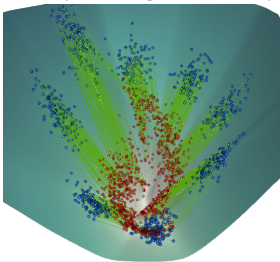
Finite-chain negatives are biased. Transport warm-up makes noise-initialised chains useful.



Transport and density shape different parts of the landscape

GLOBAL ROUTE AND LOCAL EQUILIBRIUM

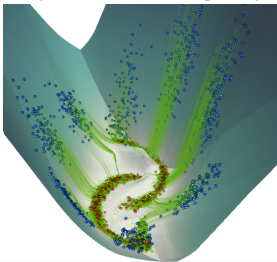
Transport alignment only



$$\mathcal{L}_{\text{OT}}$$

A coherent route from noise, but no equilibrium density constraint.

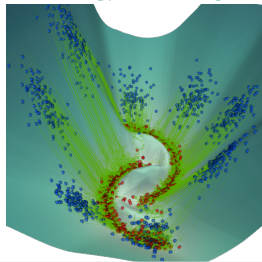
Equilibrium fitting only



$$\mathcal{L}_{\text{CL}}$$

Data-centred wells, while the off-data route remains weakly constrained.

Energy Matching



$$\mathcal{L}_{\text{OT}} + \lambda_{\text{CL}} \mathcal{L}_{\text{CL}}$$

The global route is retained while local density wells are refined.

Transport alignment teaches *how to arrive*. Equilibrium fitting teaches *where to remain*.

Sampling has an arrival phase and an equilibrium phase

DETERMINISTIC → STOCHASTIC

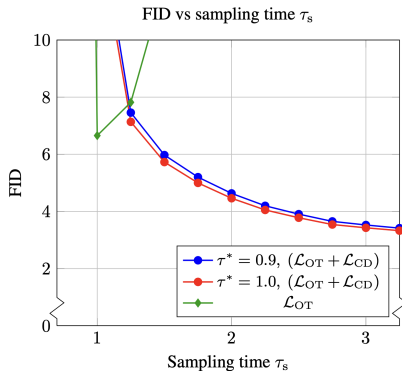
After training, sample with the same non-homogeneous Langevin dynamics:

$$X_0 \sim \rho_0, \quad dX_t = -\nabla V_\theta(X_t) dt + \sqrt{2\varepsilon(t)} dW_t.$$

noise $\xrightarrow{\varepsilon \approx 0}$ directed arrival $\rightarrow$ data region $\xrightarrow{\varepsilon = \varepsilon_{\max}}$ equilibrium

Why the second regime matters

Pure $\mathcal{L}_{OT}$ reaches the data at a calibrated time and then drifts away. The joint objective creates a distribution that can continue to equilibrate.



CIFAR-10: FID versus sampling time τ_s .

Exact equilibrium is asymptotic. Empirically FID continues improving to $\tau_s = 3.25$.

Generation closes much of the classical EBM gap

UNCONDITIONAL GENERATION

CIFAR-10 32×32, FID ↓

Model	Params	FID
Improved CD (EBM)	n/r	25.10
CLEL-large (EBM)	32M	8.61
Energy Matching	50M	3.34
OT-CFM	37M	4.04
DDPM++	62M	3.45
NCSN++	107M	2.45

ImageNet 32×32, FID ↓

Model	Params	FID
Improved CD (EBM)	n/r	32.48
CLEL-large (EBM)	32M	15.47
Energy Matching	50M	6.64
Flow Matching	196M	5.02
DDPM++	62M	8.42

In the paper's comparison, Energy Matching reduces FID by more than 50% relative to the strongest listed standalone EBM on both datasets, while remaining competitive with flow and diffusion models.

n/r = parameter count not reported in the source comparison.

The Bayesian payoff: add the task at inference

WHY EXPOSING $p(x)$ MATTERED

For measurements $y = A(x) + w$,

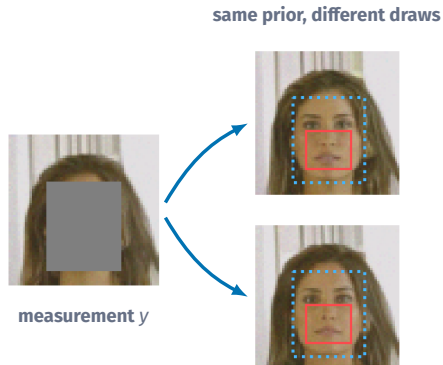
$$p(x | y) \propto \underbrace{\exp\left[-\frac{\|y - A(x)\|^2}{\zeta^2}\right]}_{p(y|x)} \underbrace{\exp\left[-\frac{V_\theta(x)}{\varepsilon_{\max}}\right]}_{p_\theta(x)}.$$

At sampler temperature $\varepsilon(t)$, simply replace V_θ by

$$U_{\theta,t}(x) = V_\theta(x) + \varepsilon(t) \frac{\|y - A(x)\|^2}{\zeta^2},$$

and run the same dynamics

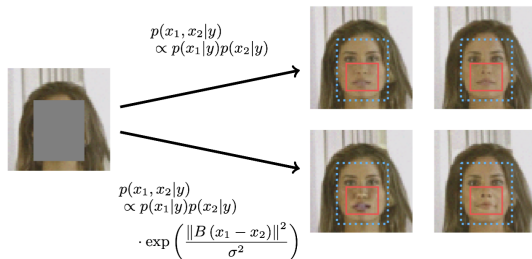
$$dX_t = -\nabla U_{\theta,t}(X_t) dt + \sqrt{2\varepsilon(t)} dW_t.$$



Learn $p_\theta(x)$ once. Change A , y , and the task energy without retraining the prior.

Interaction energies make diversity controllable

BOOST DIVERSITY WITHOUT RETRAINING

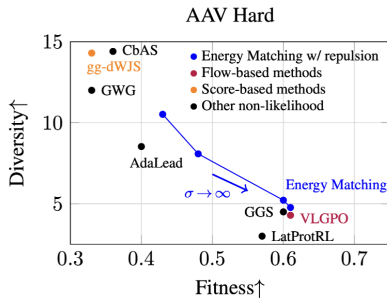
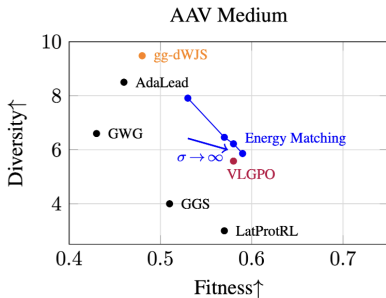


$$p(x_{1:M} | y) \propto \left[\prod_{m=1}^M p_{\theta}(x_m | y) \right] e^{-\sum_{i < j} W_{\sigma}(x_i, x_j)}, \quad W_{\sigma}(x_i, x_j) = -\frac{\|B(x_i - x_j)\|^2}{\sigma^2}.$$

$\sigma \rightarrow \infty$: independent chains $\longleftrightarrow$ smaller σ : stronger repulsion in B .
The learned single-sample prior is not retrained. The joint batch target is deliberately changed.

Repulsion improves diversity in protein inverse design

AN INFERENCE-TIME FITNESS-DIVERSITY KNOB



Fitness: predicted viral packaging efficiency

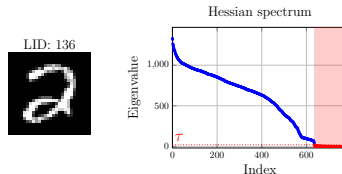
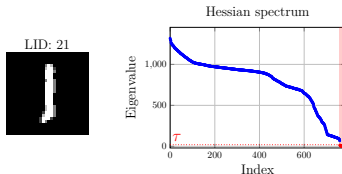
Diversity: mean pairwise sequence distance

$\sigma \rightarrow \infty$ removes repulsion.
Stronger repulsion moves along the blue trade-off curve.

One explicit energy supports generation, conditioning, and controllable batch diversity.

The Hessian reveals local intrinsic dimension

EXPLICIT CURVATURE AT THE DATA POINT



$$H_{\theta}(x) := \nabla_x^2 V_{\theta}(x) = Q \operatorname{diag}(\lambda_1, \dots, \lambda_d) Q^{\top},$$
$$\widehat{\text{LID}}_{\tau}(x) = \sum_{i=1}^d \mathbf{1}\{|\lambda_i| < \tau\} = d - \operatorname{rank}_{\tau} H_{\theta}(x).$$

Under the local-well assumption, near-zero curvature directions act as local degrees of freedom.

Why the explicit scalar helps

Energy Matching lets us differentiate V_{θ} directly at x_{data} . Diffusion-based estimates typically evaluate the score near, rather than on, the data manifold.

Spearman correlation with PNG size

MNIST 0.877

CIFAR-10 0.901

Graph Energy Matching: continuous gradients, discrete states

THE AMBIENT SPACE IS ONLY A COMPUTATIONAL DEVICE

A graph is categorical

$$x = (x_V, x_E) \in \mathcal{X} = \mathcal{X}_{\text{node}}^n \times \mathcal{X}_{\text{edge}}^{\binom{n}{2}}.$$

Its one-hot embedding supplies differentiable coordinates,

$$\hat{x} = \phi(x) \in \{0, 1\}^d \subset \mathbb{R}^d, \quad V_\theta(x) := \tilde{V}_\theta(\hat{x}),$$

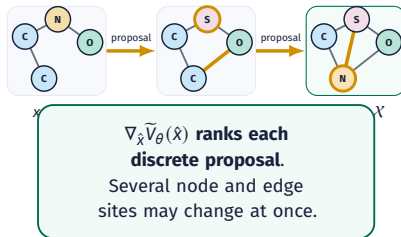
while the energy is evaluated only on valid graphs $x \in \mathcal{X}$.
Given the pair $x^{\text{data}} = T(x^0)$, interpolate every node or edge component j through a *discrete* categorical state

$$x_j^t \sim \text{Cat}\left((1-t)\phi(x_j^0) + t\phi(x_j^{\text{data}})\right), \quad x^t \in \mathcal{X}.$$

$$v = \widehat{T(x^0)} - \hat{x}^0, \quad \mathcal{L}_{\text{Flow}} = \mathbb{E}\left[\|\nabla_{\hat{x}^t} \tilde{V}_\theta(\hat{x}^t) + v\|^2\right], \\ \mathcal{L}_{\text{GEM}} = \mathcal{L}_{\text{Flow}} + \lambda_{\text{CL}} \mathcal{L}_{\text{CL}}.$$

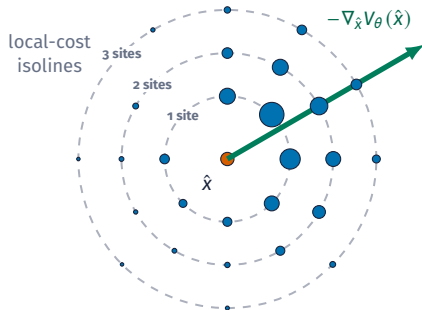
Modular: graph representation ϕ + proposal family $C(x)$ + local cost $c_{\text{loc}}(x, y)$ + invariant energy V_θ .

one proposal can change several sites



A gradient does not move a graph — it scores jumps

DISCRETE, GEOMETRY-AWARE LANGEVIN



dot area $\propto q_{\text{mix}}(x \rightarrow y)$
gradient alignment competes with jump cost
each dot is one complete graph proposal

Transport: deterministic burn-in

$$y^*(x) = \arg \min_{y \in C_N(x)} \{ \langle \nabla V_{\theta}, \hat{y} - \hat{x} \rangle + c_{\text{loc}}(x, y) \}.$$

$C_N(x)$: exactly N node/edge changes in one proposal.

Mixing: stochastic jumps + MH

$$\log q(x \rightarrow y) = -c_{\text{loc}}(x, y) - \beta^L \langle \nabla V_{\theta}, \hat{y} - \hat{x} \rangle + C,$$

$$c_{\text{loc}}(x, y) = \lambda_{\mathcal{V}}^L \|\hat{y}_{\mathcal{V}} - \hat{x}_{\mathcal{V}}\|^2 + \lambda_{\mathcal{E}}^L \|\hat{y}_{\mathcal{E}} - \hat{x}_{\mathcal{E}}\|^2.$$

MH corrects the proposal toward
 $\pi_{\theta}(x) \propto e^{-\beta_{\text{MH}} V_{\theta}(x)}.$

uniform graph noise



greedy N -site jumps



enter mixing



multi-site jumps + MH